



Network Price List

for

Standard Form Customer Connection Contract

effective

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SECTION 1: GENERAL INFORMATION

1. INTRODUCTION

In this document "we", "us", "our" and "ours" refers to Integral Energy; "you", "your" and "yours" refers to you, the customer.

Words in *italics* are explained in the Connection Contract. The Connection Contract is available at our website at www.integral.com.au. Alternatively, you can obtain a copy by calling our Customer Service staff on 131 002.

2. RATES, FEES AND CHARGES

The documents mentioned below can be found on our website.

If you have any questions in relation to network prices please contact:

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2.1 Network Price List

We have compiled this Network Price List (**Price List**) to provide you with:

- (a) a description of rates relating to our provision of *connection services* and *network use of system services*;
- (b) the basis for calculation of rates for *connection services* and *network use of system services*; and
- (c) information relating to miscellaneous fees and charges (Section 4),

when your *property* is connected to our *distribution system*.

This Price List forms part of the Connection Contract and complies with:

- the IPART Determination on NSW Electricity Distribution Pricing and associated Rules made under the National Electricity Code (June 2004); and
- the Annexure 13 "Pricing Principles" of the June 2004 NSW Electricity Distribution Pricing Determination No.2 by the IPART; and
- the *Electricity Supply Act*.

For the purposes of this Price List the following public holidays are deemed to be non-*business days*: New Year's Day, Australia Day, Good Friday, Easter Monday, Anzac Day, Queen's Birthday, Labour Day, Christmas Day, Boxing Day, and other gazetted public holidays in New South Wales.

All other non-gazetted holidays, such as bank holidays and other local holidays are deemed to be *business days*.

2.2 Network Pricing and Service Report

We have also prepared a separate document, which sets out how we determine the prices in the Price List (**Annual Network Pricing Report**). That Annual Network Pricing Report includes information on:

- (a) the level and structure of prices;
- (b) the standard of service provided;
- (c) the methodology used to derive prices and their cost basis; and
- (d) medium term directions for prices and standards of service.

3. DEFINITIONS

3.1 Meaning of words

In this Price List, unless inconsistent with the context or subject matter:

"CEO" means the Chief Executive Officer of Integral Energy or a person authorised by the Chief Executive Officer or the Board to act on that person's behalf.

"Connection Contract" means the *standard form customer connection contract* entered into between a customer and us.

"Controlled Load" has the meaning given in clause 5.3.

"DST" is Eastern Daylight Saving Time and applies for the period gazetted by the NSW Government for use as local time.

"Daily Threshold" has the meaning given in clause 9.2.

"Demand Charge" has the meaning given in clause 8.

"Distribution Loss Factor" has the meaning given in clause 10.

"EST" has the meaning given in clause 0.

"Export Energy" has the meaning given in clause 13.

"Import Energy" has the meaning given in clause 13.

"IPART" means the Independent Pricing and Regulatory Tribunal of New South Wales.

"Inclining Block Tariff" has the meaning given in clause 5.2.

"Metrology Procedures" means the New South Wales Electricity Supply Industry Metrology Procedure for types 5, 6 and 7 metering installations.

"Network Pricing and Service Report" has the meaning given in clause 2.2.

"NEM" means the National Electricity Market.

"Normal Business Hours" has the meaning given in clause 16.1.

"Price List" has the meaning given in clause 2.1.

"Pricing Category" has the meaning given in clause 5.1.

"Pricing Options" are set out in clause 5 of this Price List, only one of which will apply at any time to the *connection services* and *network use of system services* provided at each separately metered *supply point*.

"Time of Day" has the meaning given in clauses 12.1 and 12.2.

"Voltage of Supply" has the meaning given in clause 15.

SECTION 2: NETWORK PRICES

4. SECURITY

4.1 General

We:

- (a) may require you to pay *security* where the requirements set out in clauses 4.3 or 4.4 below are met; and
- (b) must repay, release or discharge the *security* to you in accordance with clause 4.5 or 4.6 below.

The amount of the *security* may be up to:

- (c) 1½ times the average quarterly account;
- (d) 1¾ times the average bi-monthly account; or
- (e) 2½ times the average monthly account.

You will not be paid interest on the *security*.

4.2 Types of security

We accept only the following types of *security*:

- (a) cash, cheque or credit card from residential or business customers; or
- (b) guarantees, including Department of Housing guarantees, from residential customers, and bankers' guarantees from business customers.

4.3 Residential customers

We may require *security* from a residential customer prior to connection only if that customer:

- (a) has left a previous supply address without settling an electricity retail bill owed to us or any other standard network distributor, the debt remains outstanding and the customer has refused and refuses to make any arrangement to pay that debt;
- (b) has been responsible for the illegal use of electricity within the previous two years; or
- (c) does not have a satisfactory credit history in our reasonable opinion or cannot demonstrate satisfactory credit history with another network distributor to our reasonable satisfaction, and we have offered the customer an instalment plan or other payment option (for example pay as you go instalments, direct debit) and the customer has refused, or failed to agree to the offer.

4.4 **Business customers**

We may require *security* from a business customer prior to connection only if the customer:

- (a) is a new business;
- (b) has been responsible for the illegal use of electricity within the previous two years; or
- (c) does not have a satisfactory credit history in our reasonable opinion or cannot demonstrate a satisfactory credit history with another network distributor to our reasonable satisfaction.

4.5 **Repayment of security paid by cash, cheque or credit card**

Security paid by cash, cheque or credit card, will be refunded when:

- (a) a residential customer has made on-time payment of bills for one year from the date of the first bill; or
- (b) a business customer has made on-time payment of bills for two years from the date of the first bill and the maintenance of a satisfactory credit rating in our reasonable opinion.

When this occurs, we will inform you, in writing, of the amount that is refundable, and will credit that amount to your account within 10 *business days*.

4.6 **Discharge or release of security paid by guarantee**

Security paid in the form of a guarantee, will be discharged or released when:

- (a) a residential customer has made on-time payment of bills for one year from the date of the first bill; or
- (b) a business customer has made on-time payment of bills for two years from the date of the first bill and has maintained a satisfactory credit rating in our reasonable opinion.

If you meet the above requirements, we will inform you that a guarantee is no longer required. We will inform you in writing, and within 10 *business days*, of the above requirements being met.

4.7 **Disconnection of supply**

If you request us to disconnect *electricity supply*, before your *security* has been refunded we will:

- (a) inform you in writing of the amount of the *security* held; and
- (b) pay the amount either to you or your account.

This must occur within 10 *business days* of you ceasing to have connection.

5. PRICING OPTIONS

5.1 Determination of Pricing Options

We assign you a **Pricing Option** from those described in clause 6 and set out in Appendix 1 and 2, when we commence supply to you under a Connection Contract with us. The assigned Pricing Option will depend on the annual energy consumption measured at the *supply point*, the supply voltage at the *supply point*, the method of connection to our *distribution system* and the type of meter(s) installed at your *property*.

If you wish us to change the Pricing Option applicable to a particular point, which is separately metered at the *property*, then you must apply to us in writing giving us 30 days notice. Any change we decide to make to the Pricing Option will take effect from the date on which your meter is next read after we process your application. We reserve the right to review your Pricing Option in the event of load change or of any other reason provided we do so acting reasonably. If we change your Pricing Option we will notify your *retail supplier*. Our CEO's decision shall be final and conclusive in determining the applicable Pricing Option.

If we agree to your request to change a Pricing Option to or from an **Controlled Load** Pricing Option more than once within a 12 month period, we may charge you or your *retail supplier* the Controlled Load conversion fee (listed in item 5 of the table in clause 16.1) for each additional change.

The available Pricing Options are listed here and described in detail below:

- Domestic;
- Controlled Load;
- Domestic Time of Use;
- General Supply Non Time of Use;
- General Supply Time of Use; and
- Demand Time of Use.

You should consider which Pricing Option or Pricing Options best suits your requirements based on the electrical load and connection voltage.

The amount that you must pay us or your *retail supplier* is calculated by multiplying the appropriate GST-inclusive "per kWh" price by the amount of electricity consumed (based on our measurement or estimate of your consumption) at each separately metered *supply point*.

This amount is then multiplied by the **Distribution Loss Factor** (see clause 10).

In addition, you must pay the relevant GST-inclusive "per day" **Network Access Charge** (see clause 7, if any) for each day from the *actual start date* until your Connection Contract with us terminates.

5.2 Domestic

A Domestic Pricing Option only applies to *connection services* and *network use-of-system services* supplied to the *supply point*, which is predominantly used for one or more of the following purposes:

- (a) private dwellings;
- (b) boarding and lodging houses, being any house in which three or more persons exclusive of the family of the proprietor thereof are lodged for hire or reward from week to week or for more than a week;
- (c) retirement villages;
- (d) residential sections of nursing homes and hospitals;
- (e) residential sections of educational institutions;
- (f) children's homes;
- (g) approved baby health centres, day nurseries and kindergartens;
- (h) churches, mosques, temples etc., being buildings or properties which are used principally for public worship or partly for public worship and partly for educational purpose; or
- (i) approved caravan sites,

and where that point has a basic meter or an interval meter that is read as a basic meter.

The Domestic single-rate structure was replaced with an **Inclining Block Tariff (IBT)** two-rate structure on 1 July 2004. Since that time the Domestic Pricing Option comprises the following components, as detailed in Appendix 1:

- a Network Access Charge (fixed component), and
- two variable components consisting of
 - 1) a **First Block Rate**, expressed on a ¢/kWh basis, to be applied to electricity consumption (kWh) up to and including 1,750 kWh per quarter; and
 - 2) a **Second Block Rate**, expressed on a ¢/kWh basis, to be applied to electricity consumption (kWh) above 1,750 kWh per quarter.

5.3 Controlled Load

A Controlled Load Pricing Option only applies where electricity load is separately metered and controlled at a *supply point* with a basic meter.

- (a) **Controlled Load 1** applies where specified appliances are controlled by our equipment so that supply may not be available between 7:00am and 10:00pm; and
- (b) **Controlled Load 2** applies where our equipment controls approved specified appliances so that electricity is available for restricted periods not exceeding 17 hours in any period of 24 hours.

Switching times will be managed to minimise network investment and meet customer needs for the load being controlled.

When a customer with Controlled Load chooses another Pricing Option, the Controlled Load meter may be removed but the Controlled Load relay must remain in place. The Controlled Load relay will remain our property, and must not be removed without the written approval of the Manager, Network Review and Compliance.

Customers with a Controlled Load relay are entitled to a controlled load network price only if all of the following conditions are met:

- (c) Controlled Load consumption is separately metered using the same type of meter as the uncontrolled portion of a customer's load;
- (d) Controlled Load consumption and uncontrolled load consumption is always synchronously read, ie on the same day; and
- (e) the Controlled Load is operated by our load control systems or time clocks.

This Pricing Option is applicable only to approved appliances as detailed in Appendix 2.

5.4 Domestic Time of Use

A Domestic TOU Pricing Option applies to a property, which is predominantly used for one or more of the purposes set out in clause 5.2 at a *supply point* with a time of use meter from which interval meter data is obtained. See also clause 6.5.

We reserve the right to assign the Domestic TOU Pricing Option to any new or existing *supply point* fitted with an interval meter.

5.5 General Supply Non Time of Use

A GS Non-TOU Pricing Option applies to low voltage electricity used for any purpose other than Domestic, at a *supply point* with a basic meter or an interval meter that is read as a basic meter. See also clause 6.5.

The General Supply Non-Time-Of-Use single-rate was replaced with an **Inclining Block Tariff (IBT)** two-rate structure on 1 July 2004. Since that time the GS Non-TOU Pricing Option comprises the following components, as detailed in Appendix 1:

- a Network Access Charge (fixed component), and
- two variable components consisting of
 - 1) a **First Block Rate**, expressed on a ¢/kWh basis, to be applied to electricity consumption (kWh) up to and including 5,250 kWh per quarter; and
 - 2) a **Second Block Rate**, expressed on a ¢/kWh basis, to be applied to electricity consumption (kWh) above 5,250 kWh per quarter.

5.6 General Supply Time of Use

A GS TOU Pricing Option applies to low voltage electricity used for any purpose other than Domestic at a *supply point* with a time of use meter from which interval meter data is obtained.

We reserve the right to assign the General Supply TOU Pricing Option to any new or existing *supply point* fitted with an interval meter.

5.7 Demand Time of Use

A Demand TOU Pricing Option applies to electricity used for any purpose that includes a Demand Charge at each *supply point*. Customers shall formally notify Integral Energy when there is an anticipated change in demand at their site, for example where plant or apparatus is being commissioned or undergoing testing, or where load is transferred or rearranged. This Pricing Option will be mandatory for all General Supply customers consuming more than 160 MWh per annum with a time-of-use meter, from 1 July 2006.

6. PRICING OPTIONS

6.1 Types of Pricing Options

Pricing Options are categories of prices, including all combinations of Pricing Options (**Combination Pricing Options**), which are available to customers depending on the type of metering equipment that is installed at a *supply point* on the customer's *property*. The Pricing Options are described in this clause 6 and are set out in Appendix 1 and 2.

Those Pricing Options apply irrespective of whether the customer is an Integral Energy regulated retail customer, an Integral Energy retail customer on a negotiated *supply contract* or a customer who has entered a negotiated *supply contract* with another licensed electricity retailer.

6.2 Standard Network Pricing Options

Standard Network Pricing Options (as set out in Table 1 of Appendix 1) are applicable to *supply points* located in the *Integral Energy distribution district*, unless one of the Pricing Options described in clauses 6.3, 6.4 or 6.5 apply.

6.3 Combination Pricing Options

Combination Pricing Options (as set out in Table 2 of Appendix 1) are combinations of Pricing Options, which are applicable to points at which a combination meter is installed. A combination meter is one which can meter both a controlled load and normal domestic (or general supply) consumption as two distinct energy flows. Combination Pricing Options are formulated based on the prices, which would ordinarily be applicable to each component of the Combination Pricing Option. For example, NC01 (Domestic/Controlled Load 1) Pricing Option consists of Domestic and Controlled Load 1 charges. Sites with Type 5 metering will be allocated different network Pricing Options (see clause 6.5).

6.4 Unmetered Supply Pricing Options

Unmetered Supply Pricing Options (as set out in Table 3 of Appendix 1) are applicable to *supply points* that are not metered. Energy consumption is calculated using the appropriate algorithm in the applicable Metrology Procedure.

6.5 Type 5 Pricing Options

Type 5 Pricing Options (as set out in Table 1 of Appendix 1) are applicable to *supply points* with a Type 5 (manually read interval) meter installed, which is considerably more sophisticated than the basic Type 6. For residential customers, the appropriate Pricing Option is N705 Domestic TOU. For all other Low Voltage customers with a Type 5 meter installed, the N845 General Supply TOU Pricing Option is the appropriate equivalent.

6.6 Site-specific Pricing Options

We may assign a Site-specific Pricing Option to any Network Customer in circumstances such as, but not limited to:

- we wish to recover investment associated with stranded or dedicated assets, which may otherwise not be recovered under a Standard Network Pricing Option,
- we agree to assign a Site-specific Pricing Option following an application from the customer, or a *retail supplier* acting on their behalf, to be assigned a Site-specific Network Pricing Option.

Applications requesting a change to a Site-specific Network Pricing Option must be submitted at least six months in advance, and the *supply point* in question must be at High Voltage or Sub-transmission level according to clause 15, and satisfy at least one of the following criteria:

- (1) Energy consumption must have been equal to or greater than 100 GWh in total for the 36 months preceding the application; or
- (2) Energy consumption must have been equal to or greater than 40 GWh per annum in each of the two financial years preceding the application; or
- (3) Monthly peak demand must have been equal to or greater than 10 MVA for 24 of the 36 months preceding the application.

If the application is accepted, a Site-specific Pricing Option will be applied to the *supply point* in question from the beginning of the financial year following the date of approval of the application.

We reserve the right to reassign a Standard Pricing Option to a *supply point* as set out in clause 6.2, and effective from the beginning of the next *billing period*, if we find that the *supply point* no longer satisfies any of the aforementioned criteria.

6.7 Embedded Generators

Any *supply point* that connects a generator to our *distribution system* must have an active NUOS account associated with it, as we will invoice a regular Network Access Charge as set out in clause 7, irrespective of whether or not an Import of Energy, as set out in clause 13, occurs at such a *supply point* during any *billing period*.

In cases where a High Voltage or Sub-transmission *supply point* exists primarily to connect a generator to our *distribution system*, and if energy consumed at that same *supply point* is less than five per cent of the energy generated during any *billing period*, then we may apply a General Supply Time Of Use Network Pricing Option, as set out in clause 5.6, to that *supply point*.

However, if the *supply point* in question exceeds the level given above for more than two months during any period of twelve months, we reserve the right to assign a Standard High Voltage or Sub-transmission Demand Time Of Use Network Pricing Option to it, effective from the beginning of the next *billing period*.

7. NETWORK ACCESS CHARGES

A Network Access Charge (NAC) is applicable to all customers. The NAC is applied as a fixed daily charge for each *supply point* to the Integral Energy *distribution system*, ie per

National Metering Identifier (NMI). More than one NAC may apply per NMI if there is more than one Pricing Option applicable to that NMI.

8. DEMAND CHARGING

8.1 Demand Metering

A Demand Charge is applicable to all customers on a Demand Pricing Option in respect of the demand for electricity their *supply points* place on the *distribution system*. Demand is treated as a component of the Data Stream of Interval Metering Data, in accordance with Section 4.1 of NEMMCO NMI Procedure ME-GN059v004.

For the purpose of this Price List, the following definitions are considered equivalent:

NEMMCO		Quadrants covered	Integral Energy	
Energy Flow Definition	NMI Master Channel Suffix		Energy (or Power) Flow Definition	Corresponding Load Angle ϕ in degrees
Import kWh	B	2, 3	Effective, generated	180°
Export kWh	E	1, 4	Effective, consumed	0°
Import kVARh	K	3, 4	Reactive, generated	Leading (Capacitive)
Export kVARh	Q	1, 2	Reactive, consumed	Lagging (Inductive)

For a particular NMI, the Demand Charge for a month is calculated based on the maximum demand (kVA) for every half-hour metering interval during that month.

Let NEEEXXXXXX be a NMI with i feeders.
 Let $E_1, \dots, E_i$ be the kWh channels for each feeder.
 Let $K_1, \dots, K_i$ be the leading kVARh channels for each feeder.
 Let $Q_1, \dots, Q_i$ be the lagging kVARh channels for each feeder.

Then the kVA for each interval (usually half an hour) is calculated as

$$kVA = m \times \sqrt{(\sum E_n)^2 + (\sum (Q_n - K_n))^2}$$

where m is the number of metering intervals in an hour (usually $m = 2$ within the **NEM**)

For *supply points* fitted with an 'EMAIL Q3' meter, the measurement of reactive energy is predefined in the configuration of the meter as the addition of lagging and leading components. Common practice is to identify this measurement as Q, in these instances the kVA for each half-hour interval is calculated as.

$$kVA = m \times \sqrt{(\sum E_n)^2 + (\sum Q_n)^2}$$

The resultant kVA for sites metered with an 'EMAIL Q3' meter is in real terms equivalent to

$$kVA = m \times \sqrt{(\sum E_n)^2 + (\sum (Q_n + K_n))^2}$$

For *supply points* covered by the Connection Contract, the difference in the above computations is not significant, as either Q or K (usually K) is nil. See clause 13 and 14 for further details.

8.2 Demand Charge Calculation

If you are on a Demand TOU Pricing Option, you must pay a monthly charge based on the maximum recorded kVA in every half hour period during the month (**Demand Charge**).

The Demand Charge is calculated as follows:

$$D_c = D_M \times d$$

where

D_c = Demand Charge for the month

D_M = the maximum demand (measured in kVA) recorded in respect of the *supply point* for the entire month

d = Demand rate (\$/kVA/month)

The demand rates used to calculate this charge are set out in Appendix 1.

8.3 Demand Pro-ration Rules

If you change your *retail supplier* part way through the month, we will calculate the Demand Charge (based on your maximum demand) for the entire month for the *supply point* and then pro-rate the charge to each *retail supplier* accordingly.

The pro-rated Demand Charge in respect of each *retail supplier* for each month is calculated as follows:

$$D_P = D_M \times d \times (t/T)$$

where

D_P = pro-rated Demand Charge in respect of the *retail supplier* for the month

D_M = the maximum demand (measured in kVA) recorded in respect of the *supply point* for the entire month

T = number of days in the entire month

t = number of days with the relevant *retail supplier* to be charged

d = Demand rate (\$/kVA/month)

Say a customer C transfers from *retail supplier* 1 (R1) to *retail supplier* 2 (R2) on the 11th of January. For C, maximum demand for the entire month was 150 kVA, and the relevant Pricing Option's demand component \$5/kVA/month.

R1 would be charged according to the following formula:

$$150 \text{ kVA} \times 5 \text{ \$/kVA} \times (10/31) = \$241.94$$

R2 would be charged according to the following formula:

$$150 \text{ kVA} \times 5 \text{ \$/kVA} \times (21/31) = \$508.06$$

9. NETWORK ACCESS CHARGE AND CONSUMPTION PRO-RATION RULES

9.1 General Consumption Pro-ratio Rules

Where Integral Energy is allowed by IPART to vary certain charges and rates as listed in Appendix 1, those variations may become effective part way through a *billing period*. The amount which each customer must pay for consumption under existing charges and rates, and for consumption under new charges and rates is calculated on a pro rata basis.

The pro-rated amount, in respect of the applicable **Network Access Charge** (NAC) and Energy Rate for each part of the *billing period* (respectively after the increase becomes effective) is calculated as follows:

$$E_P = E_M \times e \times (t/T) + n \times t$$

where

E_P = pro-rated sum of NAC and Energy Rate in respect to the *billing period*

E_M = the total consumption (measured in kWh) recorded in respect of the *supply point* for the *billing period*

T = number of days in the *billing period*

t = number of days with the relevant NAC and Energy Rate to be invoiced

e = Energy rate (¢/kWh)

n = NAC (\$/day)

Say a customer C used to have a quarterly *billing period*, and an increase in the Energy Rate becomes effective on the 31st of a 92 day *billing period*. C's energy consumption for the entire *billing period* was 920 kWh. For C, the relevant Pricing Option's NAC is 0.21 \$/day before and 0.22 \$/day after the increase, and the energy rate is 6.32 ¢/kWh before and 6.83 ¢/kWh after the increase.

For the first 30 days, C would be charged according to the following formula:

$$920 \text{ kWh} \times 6.32 \text{ ¢/kWh} \times (30/92) + 0.21 \text{ \$/day} \times 30 = \$25.26$$

For the last 62 days, C would be charged according to the following formula:

$$920 \text{ kWh} \times 6.83 \text{ ¢/kWh} \times (61/92) + 0.22 \text{ \$/day} \times 61 = \$55.08$$

9.2 Specific Consumption Pro-ratio Rules in Relation to IBT

As mentioned in clause 5.2 and 5.5, we decided to introduce IBT for both the Domestic and General Supply Non-TOU Pricing Option effective 1 July 2004. Some additional explanation is helpful for the case if a customer is on such a Non-TOU Pricing Option, but has both an interval meter installed and a monthly *billing period*.

For billing purposes, the IBT thresholds mentioned in clause 5.2 and 5.5 are converted to a daily basis (**Daily Threshold**).

If average daily consumption in the *billing period* is below the Daily Threshold:

$$E_C = E_A \times T \times P_1$$

If average daily consumption in the *billing period* is above the Daily Threshold:

$$E_C = [L_1 \times T \times P_1 + (E_A - L_1) \times T \times P_2]$$

where

E_C = consumption charge in respect to the *billing period*

E_A = average daily consumption for the *billing period* (kWh)

L_1 = Daily Threshold for the First Block (kWh)

T = number of days in the *billing period*

P_1 = Energy rate for the First Block (¢/kWh)

P_2 = Energy rate for the Second Block (¢/kWh)

Say a General Supply non-TOU customer consumed 1,800 kWh during a *billing period* of 30 days, averaging 60 kWh per day. The energy rate is 4.995 ¢/kWh for the First and 5.215 ¢/kWh for the Second Block, and the IBT threshold is 21,000 kWh per year, ie the Daily Threshold is 57.534 kWh per day.

The consumption charge is then calculated as follows:

$$57.534 \text{ kWh} \times 30 \times 4.995 \text{ ¢/kWh} + (60.0 - 57.534) \text{ kWh} \times 30 \times 5.215 \text{ ¢/kWh} = \$ 90.07$$

10. DISTRIBUTION LOSS FACTOR

The **Distribution Loss Factor** (DLF) is used by a *retail supplier* to increase the customer's metered energy amount to account for electrical losses in the *distribution system*. The DLF's shown in Appendix 1 are those current as at the date of publication of this Price List. However, they are subject to change.

The DLF's shown in this Price List do not include the transmission loss factors which are used in the market settlement of energy to take account of electrical losses in the transmission network.

Non-business Days

Off-peak: All other times.

Demand or General Supply Time of Use, Non-Business Days

Command or General Supply Point of Use, Non-Emergency Day																									
Off-Peak																									
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24

12.3 Network Billing and Daylight Saving Time

In order to maintain the same time limits during both Eastern Standard Time (EST) and Daylight Saving Time (DST), billing data is adjusted by shifting the data forward an hour to accommodate for the clock shift during DST.

This means that at the start of DST (2am on Sunday) there will be an hour of null data when the clock is shifted forwards an hour from EST to DST. Also, data for the period 23:00 to 24:00 in EST will be recorded the following day for the period 00:00 to 01:00 DST.

When DST ends, the clock will move back an hour and there will be two sets of hourly data for the period from 2am until 3am, one set generated in DST and the second set generated after the clock shift in EST. This data is aggregated for the purposes of billing the "per kWh" charge, but not for Demand charge calculations.

The table below represents how the data is shifted for DST. The value in each cell (1 to 24) is the period of the day in EST.

	00:00 - 01:00	01:00 - 02:00	02:00 - 03:00	03:00 - 04:00	(etc)	20:00 - 21:00	21:00 - 22:00	22:00 - 23:00	23:00 - 24:00
EST	1	2	3	4	(etc)	21	22	23	24
DST day 1	1	2	NULL	3	(etc)	20	21	22	23
DST	24	1	2	3	(etc)	20	21	22	23
EST day 1	24	1	2 3	4	(etc)	21	22	23	24

The first row represents a normal EST day.

The second row represents day one of DST. Note that the first two hours of the day are the same as EST. At 2am, when DST begins, the data is shifted forward one hour, resulting in a null value for the period between 2 and 3am. Following that, all data is shifted forward one hour as compared to EST.

The third row represents a normal DST day. The data from the last hour of the previous day in EST is used as the data for the first hour of the following day in DST.

The final row of the table represents the day when DST switches back to EST. The first three rows are as per normal DST days, then at 3am EST begins, which means there is a clock shift back one hour. Thus data is recorded for 2 - 3am DST and also for 2 - 3am EST. This data is aggregated for the purpose of billing the energy (per kWh) component of the network charge, but not for the Demand charge component. After 3am, data is recorded and billed as per normal for EST.

Note that while there is less total consumption during the first day of the DST period, this is made up for when the switch back to EST occurs.

13. **TREATMENT OF IMPORT/EXPORT POWER FLOWS WITHIN THE CONTEXT OF OUR NETWORK USE OF SYSTEM SERVICES CHARGES**

In the situation where an end-use customer generates into, as well as consumes energy from, Integral Energy's *distribution system*, *network use of system services* charges apply to the energy consumed by the customer (**Export Energy**). The energy generated back into Integral Energy's *distribution system* (**Import Energy**) is not recognised for *network use of system services* purposes, unless it is covered under a specific agreement.

Network use of system services charges are based on Export Energy only, ie Import Energy will not be subtracted from the Export Energy.

This policy also extends to Demand Charges, ie **Import Demand** will not be subtracted from **Export Demand** when calculating *network use of system services* charges.

Furthermore, metering shall be configured so that reactive energy is measured only when associated with energy consumed by the customer (Export Energy).

14. **NATIONAL METERING IDENTIFIER (NMI)**

We issue a National Metering Identifier (NMI) for each site in accordance with Section 1.7 of NEMMCO NMI Procedure ME-GN1059. As a general rule, each NMI will attract one network charge, ie one charge for network access and a coincident Demand charge if there are multiple meters at the NMI site. There will be some exceptions to this rule in special circumstances (eg: customers with backup supplies), however they will be dealt with on a case by case basis.

15. **VOLTAGES OF SUPPLY**

We reserve the right to determine the voltage of supply for a particular customer based on the size and nature of the load to be connected. Voltage levels referred to in the prices are:

- (a) Low Voltage (LV) - nominally 240 / 415 V;
- (b) High Voltage (HV) - nominally 12.7 kV SWER, 11 or 22 kV; and
- (c) Sub-Transmission (ST) - 33, 66 or 132 kV.

SECTION 4: MISCELLANEOUS FEES AND CHARGES

16. CHARGES FOR MISCELLANEOUS SERVICES

In accordance with the June 2004 NSW Electricity Distribution Pricing Determination No.2 by the IPART, Distribution Network Service Providers can levy various charges for services relating indirectly to the conveyance of electricity.

We will levy charges for these services on customers or their *retail suppliers*. If we levy the *retail supplier* then the *retail supplier* must decide whether to absorb the charge or pass it through to the end use customer.

16.1 Application Of Miscellaneous Fees And Charges

Unless otherwise specified, the fees and charges set out below apply between 7.30am and 4pm on *business days* (**Normal Business Hours**). Additional costs may be incurred for work undertaken at the specific request of customers outside these times.

The CEO has the authority to waive any of the fees set out below in individual cases, dependent upon the circumstances.

Miscellaneous Fees and Charges

Description of Fee or Charge (including Rate Code)	Excl. GST	Incl. GST	Rate Code
1. Disconnection/Reconnection Visit (a) Where it is necessary for us to call at a customer's <i>property</i> for the purpose of disconnecting supply for breach by the customer of a customer supply contract or a customer connection contract, where the disconnection does not occur on that occasion (for example, because of non-payment of an amount due under the contract or non-payment or provision of a <i>security</i>, but we do not proceed with disconnection because the customer tenders acceptable payment or <i>security</i>), the personal visit fee is:	\$35.00	\$38.50	CDH3
(b) Where it is necessary to call at a customer's <i>property</i> for the purpose of disconnecting supply for breach by the customer of a customer supply contract or a customer connection contract, or where a retail supplier has requested that the supply to the customer be disconnected, following the procedures laid out in our <i>Standard Form Customer Connection Contract</i> (for example, because of non-payment of an amount due under the contract or non-payment or provision of a <i>security</i>, and the customer does not tender acceptable payment or security and disconnection occurs), a fee is payable as follows: (i) Disconnection, reconnection at meter box: (ii) Disconnection, reconnection at pole top or pillar box: (Disconnection will occur at the pole top or pillar box where access is denied for disconnection, or where there is evidence that the customer has reconnected supply in breach of the <i>electricity laws</i> or of a customer supply contract, or a customer connection contract.) (iii) Reconnection outside normal business hours. (For safety reasons, a reconnection outside normal business hours is only available at a meter box.)	\$70.00 \$117.00 \$75.00	\$77.00 \$128.70 \$82.50	CDF3 DS18 AC02
2. Special Meter Reading Fee Where both the following apply, a charge for a special meter reading may be levied:	\$35.00	\$38.50	AM01, -02, -04

Description of Fee or Charge (including Rate Code)	Excl. GST	Incl. GST	Rate Code
(a) where a "special meter read" as defined in the <i>Market Operations Rule (NSW Rules for Electricity Metering) No. 3 of 2001</i> takes place (but excluding any special meter reading of Metering Installations Types 1-4, which is an Excluded Distribution Service); and (b) where physical access to a meter on a customer's premises has been sought on an occasion, and when access was sought on that occasion physical access to the meter could not be obtained. We will not levy a special meter reading charge in the following circumstances: (a) if a customer requests a special meter reading to verify the accuracy of an original meter reading undertaken by us and that the special meter reading confirms that the original meter reading was inaccurate; or (b) for a meter reading undertaken by us for the purpose of issuing a final account when the customer is vacating the <i>property</i>.			
3. Meter Test Fee Fee for the testing of a meter at the customer's request: We will not be required to test the meter if the customer refuses to pay us this fee in advance. If the meter is found to be inaccurate, we will refund any amount paid in advance and the customer will not be liable to pay this fee.	\$58.00	\$63.80	MT01
4. Conveyancing Inquiry Applies to the supply of information regarding the availability of electricity supply, the presence of our equipment, power lines etc for property conveyancing. Desk inquiry fee: Field visit fee: * GST is not applicable	\$29.00 \$58.00	\$29.00* \$58.00*	none none

Description of Fee or Charge (including Rate Code)	Excl. GST	Incl. GST	Rate Code
5. Controlled Load Conversion Fee (a) For the first change to or from an <i>Controlled Load Pricing Option</i> within a 12 month period the fee is: (b) For each additional change to or from an <i>Controlled Load Pricing Option</i> within the same 12 month period the fee is:	Free \$47.00	Free \$51.70	none DM02
6. Rectification of Illegal Connection Fee Applies for work to rectify wiring, work or connections which are unauthorised or which are in breach of the <i>electricity laws</i> or the customer's contract with us, and to restore service to our requirements. Fee to rectify illegal connections:	 \$175.00	 \$192.50	 CI03

Please note:

Other fees and charges may also be applicable under special circumstances. For example, there are fees and charges relating to contestable work, *connection services* and asset relocations. Please contact us on 131 002, or visit our web site www.integral.com.au.

APPENDIX 1: INTEGRAL ENERGY NETWORK PRICING OPTIONS

Table 1 - Standard Network Pricing Options

Standard Network Pricing Options (GST Inclusive & Exclusive) effective 1 July 2005

Integral Energy Standard Network Prices	Code	Network Access Charge		Energy Rate												Demand Rate		Distribution Loss Factor	DLF Code
				Non-Time Of Use, Block 1 ¢ / kWh		Non-Time Of Use, Block 2 ¢ / kWh		Time Of Use Peak ¢ / kWh		Time Of Use Shoulder ¢ / kWh		Time Of Use Off-Peak ¢ / kWh							
		\$ / day		Excl. GST	Incl. GST	Excl. GST	Incl. GST	Excl. GST	Incl. GST	Excl. GST	Incl. GST	Excl. GST	Incl. GST	Excl. GST	Incl. GST	Excl. GST	Incl. GST		
Domestic (IBT)	N70	0.190	0.2090	5.423	5.9653	5.910	6.5010										1.0855	HLVL	
Controlled Load 1 (Off-Peak 1)	N50	0.020	0.0220	0.417	0.4587												1.0855	HLVL	
Controlled Load 2 (Off-Peak 2)	N54	0.020	0.0220	1.704	1.8744												1.0855	HLVL	
Domestic, Type 5 Meter	N705	0.470	0.5170					7.375	8.1125	6.940	7.6340	2.290	2.5190				1.0855	HLVL	
Domestic, Type 6 Meter	N706	0.210	0.2310					7.375	8.1125	6.940	7.6340	2.290	2.5190				1.0855	HLVL	
General Supply Non-TOU (IBT) *	N90	0.190	0.2090	4.680	5.1480	5.084	5.5924										1.0855	HLVL	
General Supply TOU *	N84	0.300	0.3300					6.928	7.6208	6.014	6.6154	1.589	1.7479				1.0855	HLVL	
General Supply TOU, Type 5 M. *	N845	0.510	0.5610					7.696	8.4656	7.336	8.0696	2.356	2.5916				1.0855	HLVL	
LV TOU Demand *	N19	5.140	5.6540					1.739	1.9129	1.526	1.6786	0.360	0.3960	6.809	7.4899		1.0855	HLVL	
HV TOU Demand #	N29	8.520	9.3720					1.285	1.4135	1.158	1.2738	0.320	0.3520	4.936	5.4296		1.0360	HHVL	
ST TOU Demand	N39	12.330	13.5630					1.261	1.3871	1.134	1.2474	0.303	0.3333	3.754	4.1294		1.0182	HSTL	

* If customer is directly connected to a distribution substation: Distribution Loss Factor = 1.0702 (HLVT)

If customer is directly connected to zone substation 11/22 kV busbar: Distribution Loss Factor = 1.0227 (HHVT)

The Distribution Loss Factors and Values given in this table are effective from 00.00 h EST, 01 July 2005, and are subject to change.

All energy rates - with the exception of Controlled Load 1 & 2 - include recovery of the Energy Savings Fund contribution.

Table 2 - Combination Pricing Options

Combination Pricing Options - Network Prices (GST Inclusive & Exclusive)

effective 1 July 2005

(Applicable to Customers who have a combination meter - Please refer to clause 6.3.)

Integral Energy Standard Network Prices	Code	Network Access Charge		Energy Rate						Distribution Loss Factor	DLF Code
				Uncontrolled Non-Time Of Use, Block 1		Uncontrolled Non-Time Of Use, Block 2		Controlled Non-Time Of Use			
		\$ / day		¢ / kWh		¢ / kWh		¢ / kWh			
		Excl. GST	Incl. GST	Excl. GST	Incl. GST	Excl. GST	Incl. GST	Excl. GST	Incl. GST		
Domestic (IBT) + Controlled Load 1	NC01	0.210	0.2310	5.423	5.9653	5.910	6.5010	0.417	0.4587	1.0855	HLV
Domestic (IBT) + Controlled Load 2	NC02	0.210	0.2310	5.423	5.9653	5.910	6.5010	1.704	1.8744	1.0855	HLV
General Supply IBT + Controlled Load 1 *	NC03	0.210	0.2310	4.680	5.1480	5.084	5.5924	0.417	0.4587	1.0855	HLV
General Supply IBT + Controlled Load 2 *	NC04	0.210	0.2310	4.680	5.1480	5.084	5.5924	1.704	1.8744	1.0855	HLV

* If customer is directly connected to a distribution substation: Distribution Loss Factor = 1.0702 (HLVT)

The Distribution Loss Factors and Values given in this table are effective from 00.00 h EST, 01 July 2005, and are subject to change.

All energy rates - with the exception of Controlled Load 1 & 2 - include recovery of the Energy Savings Fund contribution.

Table 3 - Unmetered Supply Pricing Options

Unmetered Supply Pricing Options
(GST Inclusive & Exclusive) effective 1 July 2005

(Street Lighting Pricing Options applicable to Public Lighting installations only.)

Integral Energy Standard Network Prices	Code	Network Access Charge \$ / day		Energy Rate ¢ / kWh		Distribution Loss Factor	DLF Code
		Excl. GST	Incl. GST	Excl. GST	Incl. GST		
Street Lighting	ENSL	0.000	0.0000	4.608	5.0688	1.0855	HLVL
Nightwatch	ENNW	0.000	0.0000	2.959	3.2549	1.0855	HLVL
Other unmetered supplies		Price on Application					

The Distribution Loss Factors and Values given in this table are effective from 00.00 h EST, 01 July 2005, and are subject to change.

All energy rates - with the exception of Controlled Load 1 & 2 - include recovery of the Energy Savings Fund contribution.

APPENDIX 2: CONTROLLED LOAD APPLIANCES, TERMS AND CONDITIONS

Controlled Load 1

The Controlled Load 1 Pricing Option applies where specified appliances approved by the CEO are controlled by our equipment (see clause 5.3 of the Price List), so that supply may not be available between 7 am and 10 PM EST. Supply will be made available for selected periods between 10 PM and 7 am EST, as approved by the CEO.

Storage Water Heaters

In relation to a heating unit in a storage water heater, the following additional conditions must all be met:

- (a) the rated hot water delivery of the storage water heater is not less than 100 litres, unless otherwise approved by the CEO;
- (b) our equipment controls the supply of electricity to the heating unit in the storage heaters;
- (c) the operation of any booster heating units is controlled in such a way that simultaneous operation with the main heating units is not possible; and
- (d) unless otherwise approved by the CEO, heating units must be arranged as multiples of 4.8 kW in accordance with the following table:

Rated Hot Water Delivery (in Litres)	Number and Rating of Heating Elements
Up to and including 400	1 x 4.8 kW
Above 400 and not exceeding 630	2 x 4.8 kW
Above 630	As necessary to provide the full amount of heat in approximately 8 hours but in any case not more than 20 watts/litre rated hot water delivery.

NOTE: The above requirements may be varied where a Controlled Load element is provided as a booster for a solar water heater. Controlled Load elements are available to Domestic and General Supply small retail customers.

Other Appliances

In relation to swimming pool pumps, pool heating equipment, dishwashers, clothes dryers, washing machines and other appliances approved by the CEO (other than those described above) the following additional conditions must all be met:

- (a) an approved storage water heater or storage space heater is also installed and supplied at the same Controlled Load 1 Pricing Option;
- (b) each appliance is permanently connected to the fixed wiring;

- (c) all Controlled Load circuits originate at the meter board and are controlled by our equipment so that supply is available during specified Controlled Load hours; and
- (d) for pool heating, the equipment rating shall not exceed 520 watts per square metre of the water surface, unless approved by the CEO.

Noise Control

Local councils may impose conditions relating to the use or operation of equipment causing offensive noise. Air conditioners, swimming pool pumps and heat pump motors may be subject to such conditions and customers should consult the local council before arranging for such equipment to operate at night on Controlled Load.

Thermal Storage Space Heaters (Heat Banks)

In relation to thermal storage space heaters, the following additional conditions must be met:

- (a) the thermal storage space heaters are controlled by our equipment and the aggregate rating is not less than 3 kW, and
- (b) the property must utilise the Domestic Pricing Option.

Under Floor Heating

In relation to under floor heating, the following additional conditions must be met:

- (a) the under floor heaters are controlled by our equipment and the aggregate rating is not less than 3 kW, and
- (b) the property must utilise the Domestic Pricing Option.

Ice Storage Systems

Controlled Load 1 also applies to ice storage systems provided the unit is controlled by our equipment and the aggregate rating is not less than 3 kW.

Transfer Between Pricing Options

A switch that transfers equipment normally supplied as a Controlled Load to another Pricing Option is not permissible.

Existing Installations

Storage water heaters and thermal storage space heaters, previously approved for connection as a Controlled Load, will continue to be eligible for supply under the Controlled Load 1 Pricing Option.

Special conditions - single person and dual occupant aged person accommodation

Notwithstanding the rated hot water delivery requirements of the Controlled Load 1 Pricing Option, in the case of single and dual occupant aged person accommodation owned and controlled by the NSW Department of Housing, or some institution/charity as defined by the CEO, the minimum rated hot water delivery may be reduced in accordance with the following table:

Number of Occupants in Property	Minimum Hot Water Delivery Rating	Minimum Kilowatt Rating
1	80 litres	3.6 kW
2	125 litres	3.6 kW

Controlled Load 2

The Controlled Load 2 Pricing Option applies where specified appliances approved by the CEO are controlled by our equipment (see clause 5.3 of the Price List), so that electricity is available for restricted periods not exceeding 17 hours in any period of 24 hours.

The same terms, conditions and restrictions as listed for Controlled Load 1 are applicable for Controlled Load 2, with the following exceptions:

- a) The Controlled Load 2 Pricing Option can be applied to an electric heat pump with a minimum tank size of 250 litres, but that pump cannot be consequently transferred to the Controlled Load 1 Pricing Option.
- b) The Controlled Load 2 Pricing Option does not apply to Ice Storage Systems.
- c) Special conditions - single person and dual occupant aged person accommodation - the following table replaces the one applicable to Controlled Load 1:

Number of Occupants in Property	Minimum Hot Water Delivery Rating	Minimum Kilowatt Rating
1 or 2	80 litres	3.6 kW